

EPFL — Fall 2026

COM-309

Introduction to quantum information processing

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This document was originally created for my own use, but I am sharing it because I believe it may be useful to others for studying and reference purposes. However mistake can happen and there will inevitably be some errors, if you found any, you are more than welcome to send me a message (Arthur Hbe on telegram) or to directly do a PR on the repo

<https://github.com/Arthur926564/LectureNotes>

If you think that something is missing or that something is not clear, I will be more than happy to add your modification to this document.

Unless otherwise stated, all material in these notes is based on the course content, including the diagrams, explanations, and examples.

Note also that these notes were taken during the lecture, but not only; some materials was added later (that's why you may read remarks that sounds like I knew the course before taking it).

Finally, thanks a lot to Joachim Favre for his help with the setup and whose style is used in this document.

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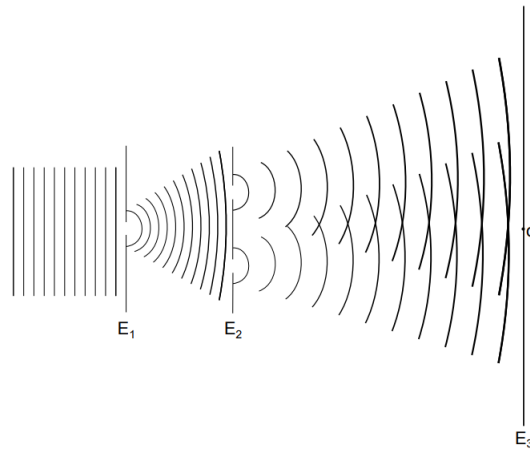
Introduction

Overview of the course	Quantum information processing is a sort of unification of the two followings fields: physics and computer science/ communication . Quantum physics is a new field of physics that emerged around 1900/1930, we found that that the behaviour of particle at a really small scale was really different of what we expected, uncertainty was the new discovery at atomic scale. We are not able to exactly predict how a particle behave even by knowing every information that could possibly be known.
Documents	The class materials are typed into lecture notes which covers this course and the quantum computation course CS-308 (chapters 9, 10, 11, 12, 13, 14).

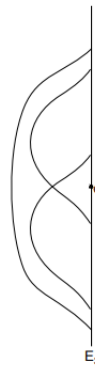
Chapter 1

Wave-Particle Duality

Interference in Double slit exp.



What this has shown is that light behaved like a wave in our case. If light were in fact particle, the result of the following experiences would be different. The resulting wave would be something like this:



However lights behaving like a wave is something that we can all agree on, on the other case, imagine doing the same experiment but for electrons or C_{60} , the resulting frequency would corresponds to a wave as for lights. And this frequency is **constant** i.e., we can send electrons one after the other then after collecting the statistics; remarkably the function we get is the same prediction as the one of wave.

Chapter 2

Mathematical notions

2.0.1 Recap of linear algebra in Dirac notation

The arena for quantum mechanics is Hilbert space (a vector space on $\mathbb{C}$).

Notation

The language in order to speak in quantum science is **Dirac** notation.

Definition 1. Ket

$$\vec{\phi} = |\phi\rangle = \begin{pmatrix} \phi_1 \\ \phi_1 \\ \vdots \\ \phi_n \end{pmatrix}, \phi_i \in \mathbb{C}$$

Definition 2. Bra

$$\vec{\phi}^\dagger = \langle\phi| = (\phi_1^*, \phi_2^*, \dots, \phi_D^*), \phi_i \in \mathbb{C}$$

Kets and Bras are the foundation of the language of quantum science, those will be used extensively throughout the whole course.

Definition 3. Inner product

In our new space, we will define the inner product as

$$\vec{\phi}^\dagger \cdot \vec{\phi} = \sum_{i=1}^D \phi_i^* \cdot \phi_i$$

Properties

As for vector in classical vector space, vector inside Hilbert space also end up having properties that we can use in order to prove theorems:

$$\begin{aligned} \langle\phi||\varphi\rangle^* &= \langle\varphi||\phi\rangle \\ (\alpha^* \langle\phi_1| + \beta^* \langle\phi_2|)|\varphi\rangle &= \alpha^* \langle\phi_1||\varphi\rangle + \beta^* \langle\phi_2||\varphi\rangle \end{aligned}$$

The positivity of bracket:

$$\langle\phi||\phi\rangle \geq 0$$

We also have triangle inequality and Cauchy-Schwartz:

$$\begin{aligned} \|\phi + \varphi\| &\leq \|\phi\| + \|\varphi\| \\ |\langle\phi||\varphi\rangle| &\leq \|\phi\| \|\varphi\| \end{aligned}$$

Orthonormal basis in $\mathcal{H}$

the dimension of our space is denoted as $\dim \mathcal{H} = D$. The basis of that space is composed of basis vector which are noted as:

$$|u_1\rangle, |u_2\rangle, \dots, |u_D\rangle$$

Where

$$\langle u_i | u_j \rangle = \begin{cases} 1 & i = j \\ 0 & j \neq i \end{cases}$$

as vector that are not equal are orthogonal hence their inner product equals 0.

Example

The most common space that will be used during this course is $\mathbb{C}^2 = \left\{ \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \alpha, \beta \in \mathbb{C} \right\}$. In this case we define our vector as

$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Tensor product

The goal of a tensor product is to combine spaces together.

Definition 4. Given $\mathcal{H}_1$ and $\mathcal{H}_2$ we define $\mathcal{H}_1 \otimes \mathcal{H}_2$ as

$$|u_1\rangle, |u_2\rangle, \dots, |u_{D_1}\rangle \otimes |v_1\rangle, |v_2\rangle, \dots, |v_{D_2}\rangle \rightarrow |u_i\rangle \otimes |v_j\rangle = |u_i, v_j\rangle$$

Where $i = 1, \dots, D_1$ and $j = 1, \dots, D_2$.

Example

Taking again $\mathbb{C}^2$ We want to convert it into $\mathbb{C}^2 \otimes \mathbb{C}^2$, the new basis of our space this gives:

$$|0\rangle \otimes |0\rangle = |0\ 0\rangle$$

$$|0\rangle \otimes |1\rangle = |0\ 1\rangle$$

$$|1\rangle \otimes |0\rangle = |1\ 0\rangle$$

$$|1\rangle \otimes |1\rangle = |1\ 1\rangle$$

This allows us to have multiple qbits inside one state, which later on will allow us to have the notion of entanglement.

If we wanted to translate this into classical linear algebra, this result for the tensor product would be the following: given two vectors $\vec{v} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $\vec{u} = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ the tensor product would be computed as

$$\vec{v} \otimes \vec{u} = \begin{pmatrix} \alpha \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \\ \beta \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}$$

Inner product

in $\mathcal{H}_1 \otimes \mathcal{H}_2$

$$(\langle \phi_1 | \otimes \langle \phi_2 |)(| \phi_1 \rangle \otimes | \phi_2 \rangle) = \langle \phi_1 | \phi_1 \rangle \langle \phi_2 | \phi_2 \rangle$$

Matrix

Definition 5. A matrix $A : \mathcal{H} \rightarrow \mathcal{H}$ is an operator defined in a given an orthonormal basis $\{|1\rangle, |2\rangle, \dots, |D\rangle\}$ which can be represented as an $D \times D$ array where $A_{ij} = \langle i | A | j \rangle$. We can even define our matrix as:

$$A = \sum_{i,j=1}^D A_{ij} |i\rangle \langle j|$$

Where $|i\rangle \langle j|$ is the matrix that has a 1 at the position (i, j) and 0 everywhere else.

*Tensor product
of matrices*

Given two Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ we know how to compute the inner product space of the tensor product between our two spaces $\mathcal{H}_1, \mathcal{H}_2$ as seen before. What we want to see now is the tensor product between two matrices A and B .

To do so we'll use the definition of the matrix using the 'ket bra' notation of the definition above.

$$A \otimes B = \sum_{i,j,l} A_{ij} B_{jl} \underbrace{|i\rangle\langle j| \otimes |k\rangle\langle l|}_{(|i\rangle \otimes |k\rangle)(\langle j| \otimes \langle l|)}$$

Remark 1. The notation $|i\rangle \otimes |k\rangle$ can be simplified by $|i, k\rangle$.

$$A \otimes B = \sum_{i,j=1}^{D_1} \sum_{k,l}^{D_2} A_{ijkl} |i, k\rangle\langle j, l|$$

The resulting dimension of our matrix is given by $D_1 D_2 \times D_1 D_2$ where

$$(A \otimes B)_{ik,jl} = A_{ij} B_{kl}$$

Example

Given the product of $\mathbb{C}^2 \otimes \mathbb{C}^2$, let us see what would be the tensor product of the two matrices:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

The goal is to compute $A \otimes B$.

$$\begin{aligned} A \otimes B &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ &= \begin{pmatrix} a \cdot B & bB \\ cB & dB \end{pmatrix} \\ &= \begin{pmatrix} ae & af & be & bf \\ ad & ah & bd & bh \\ ce & cf & ge & gf \\ cd & ch & gd & gh \end{pmatrix} \end{aligned}$$

2.0.2 Hermitian matrix

First let us do some notation:

$$\begin{aligned} A^\dagger &= \sum_{i,j} A_{ij}^* |j\rangle\langle i| \\ &= \sum_{i,j} A_{ij}^* |i\rangle\langle j| \end{aligned}$$

We have the following:

$$(A^*)_{ij} = A_{ij}^*$$

**Hermitian
matrices**

Definition 6. An Hermitian matrix is a given operator A on an Hilbert space $\mathcal{H}$ where

$$A = A^\dagger$$

Unitary matrices

Definition 7. An unitary matrix is an Hermitian operator U on an Hilbert space $\mathcal{H}$ where

$$U^{-1} = U^\dagger$$

Here an equivalent property can be derived

$$UU^\dagger = U^\dagger U = I$$

Which is equivalent to say for two any given pure states $|\phi\rangle, |\varphi\rangle$

$$\langle\phi||\varphi\rangle = \langle\phi|UU^\dagger|\varphi\rangle = (U|\phi\rangle)^\dagger \cdot (U|\varphi\rangle)$$

Spectral theorem
for Hermitian
matrices

Theorem 1. Given a Hermitian matrices A we have that all eigenvalues of our matrix are real number where their respective eigenvector are orthonormal.

$$A|u_i\rangle = \lambda_i|u_i\rangle$$

with $\lambda_i \in \mathbb{R}$.

In particular, we have:

$$A = \sum_{i=1}^d \lambda_i |v_i\rangle\langle v_i|$$

2.1 References

Bibliographie

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